Estimation of Unobservable Household Time-Use Decision-Making Parameters

UNC COLLEGE OF ARTS AND SCIENCES Economics

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Introduction

- Quantitative models allow **predictions of future**
- Method of Simulated Moments (MSM) can help **build these models** if key variables are unobserved
- Del Boca and Flinn (DBF) use 2005 Panel Study of Income Dynamics (PSID) data to **model time use choices for household members** (2012)

GOALS

1. **Replicatio**n: Replicate DBF
2. **Sample Robustness:** Variation over samples
3. **Initialization Robustness:** Variation over 1st guesses
4. **Novel Years:** Estimate novel year data

Empirical Methods

Plan for Each Goal:

a. Replication: Reconstruct Data and Initial Θ

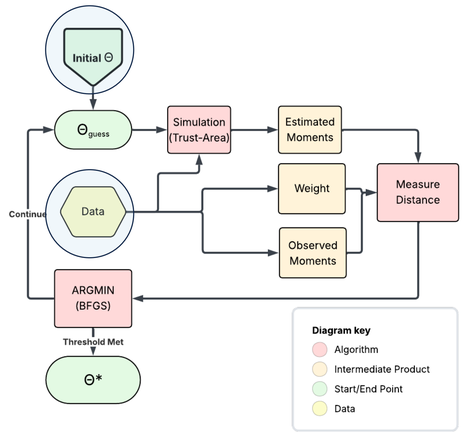
b. Sample Robustness: Change Initial Θ

c. Initialization Robustness & Novel Years: Change Data

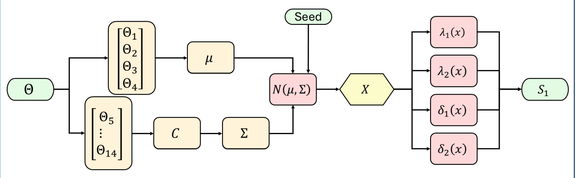
Bootstrap Estimates

- Replication: draw 1,000,000, resample 10,000x
- Else: draw 1,000, resample 100x

Method of Simulated Moments



Estimate Conversion



Results

Replication Results: Means and S.E.

	Present Estimates Estimate	S.E.	Del Boca and Flinn, 2012 Estimate	S.E.
$\mu(\lambda_1)$	0.499**	1.1e-04	0.469	0.012
$\sigma(\lambda_1)$	0.113	7.3e-05	0.152	0.033
$\mu(\lambda_2)$	0.481***	1.3e-04	0.419	0.023
$\sigma(\lambda_2)$	0.132***	8.3e-05	0.233	0.022
$\mu(\delta_1)$	0.061	1.8e-05	0.064	0.004
$\sigma(\delta_1)$	0.018	1.5e-05	0.033	0.011
$\mu(\delta_2)$	0.109	3.4e-05	0.113	0.004
$\sigma(\delta_2)$	0.034	3.0e-05	0.029	0.008

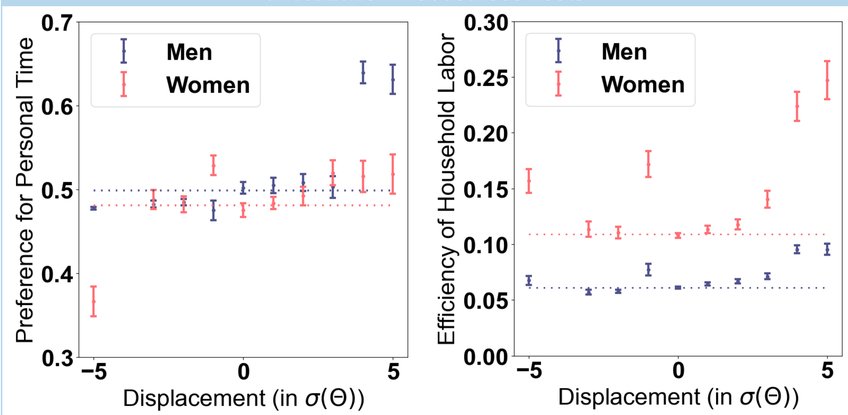
- **Close replication** of DBF
- Differences in leisure preferences
  - Likely from difference in total allowable time

Sample Robustness Test

Characteristic	Sample-Robust Estimate	S.E.	Del Boca and Flinn, 2012 Estimate	S.E.
$\mu(\lambda_1)$	0.501	0.020	0.469	0.012
$\sigma(\lambda_1)$	0.125	0.030	0.152	0.033
$\mu(\lambda_2)$	0.468	0.015	0.419***	0.023
$\sigma(\lambda_2)$	0.123	0.040	0.233***	0.022
$\mu(\delta_1)$	0.061	0.004	0.064	0.004
$\sigma(\delta_1)$	0.022	0.010	0.033	0.011
$\mu(\delta_2)$	0.109	0.009	0.113	0.004
$\sigma(\delta_2)$	0.045	0.022	0.029	0.008

- **Robust to sample changes**
  - Low S.E., especially on μ's
- Sample-related variance in estimates **may explain difference** in men's preferences

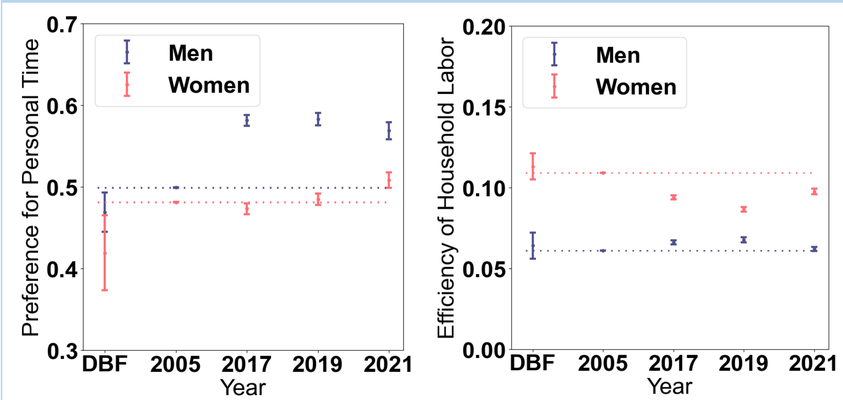
Initialization Robustness Tests



- **Some robustness** to initialization: similar preferences within 3σ

$\Delta_\sigma = -4$ did not converge

Novel Year Estimates



- Preference changes for men, efficiency **change for men and women**

2023 did not converge

Data

- Years: 2005, 2017, 2019, 2021, & 2023 PSID
- Criteria: different-sex, married, dual-income, ≤2 kids

Based on the limited demographics in DBF, our **samples do not significantly differ**

Comparison of Sample Demographics

	My Data (2005) Men	Women	DBF Men	Women
time: labor	43.589	33.159	45.706	38.588
	(12.110)	(13.498)	(8.546)	(10.512)
time: house	7.994	15.763	7.787	14.920
	(6.741)	(10.557)	(6.418)	(9.428)
wage	25.130	19.271	22.009	15.823
	(18.706)	(14.835)	(13.626)	(9.327)
nonlabor income	95.618		118.151	
	(62.061)		(182.526)	

Theoretical Model

Each household maximizes the average utility:

- Wealth is used communally
- Time spent to help self or both partners

$$W_i = 0.5u_{i,1}\left(t_{i,1}, t_{i,2}, \tau_{i,1}, \tau_{i,2}\right) + 0.5u_{i,2}\left(t_{i,1}, t_{i,2}, \tau_{i,1}, \tau_{i,2}\right)$$

$$u_{i,j} = \lambda_{i,j}\ln(T - t_{i,j} - \tau_{i,j}) + (1 - \lambda_{i,j})\ln K_i$$

$$K_i = \tau_{i,1}^{\delta_{i,1}}\tau_{i,2}^{\delta_{i,2}}M^{1-\delta_{i,1}-\delta_{i,2}}$$

$$M_i = w_{i,1}h_{i,1} + w_{i,2}h_{i,2} + Y_i$$

where,

$$T_{i,j} = l_{i,j} + t_{i,j} + \tau_{i,j}$$

Discussion

- **Replication: successful!**
- **Robustness: moderate** to sample and initialization
- Compared to 2005, in recent years:
  - Men prefer **personal time more**
  - Women and men **more similar in efficiency**

Future Directions

- Model realism (e.g. multiple public goods, split income)
- Include **same-sex households**
- Allow **power imbalances** (e.g. one partner's preferences weigh more)
- Evaluate quantitative implications

References